

BIOINFORMATICS HACKATHON · 2026

Everything OMOP

Federated learning on harmonised biobank data

SUBPROJECT 1

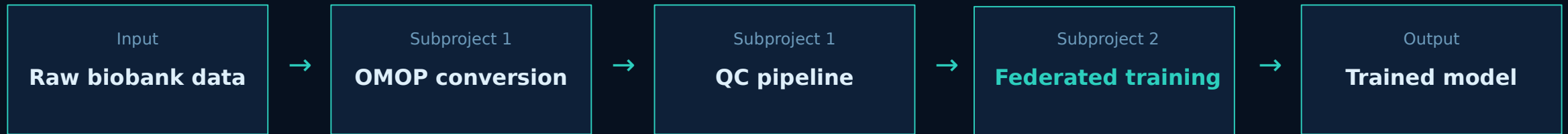
Automated biobank → OMOP conversion and quality control

SUBPROJECT 2

Proof-of-concept federated learning on OMOP datasets

OVERVIEW

End-to-end pipeline



SUBPROJECT 1

OMOP conversion & quality control

CONVERSION PIPELINE

- Identify minimal required tables and fields common across source biobank schemas
- Map source concepts to OHDSI standard vocabularies using Usagi, with manual review
- Framework is extensible to arbitrary biobank schemas — no source-specific engineering required

QUALITY CONTROL

- Completeness, date logic, unit standardisation, and concept mapping coverage
- Evaluated comparatively against a Synthea-derived reference OMOP dataset

SUBPROJECT 2

Federated learning on OMOP data

IMPLEMENTATION

- Orchestrated with [NVFlare](#) (NVIDIA Federated Learning Application Runtime Environment)
- Custom OMOP dataloader for standardised ingestion across simulated sites
- Federated Averaging ([FedAvg](#)) baseline training strategy

SCOPE & OBJECTIVE

- Simple linear model trained across simulated federated sites
- Goal is [end-to-end pipeline functionality](#), not predictive performance
- Directly consumes OMOP output from Subproject 1